

Where we are

Almost finished A3 grading (checking missing ones & Slack days)

Final project proposals: all graded (detailed feedback in two weeks)

Three more **lectures** by me: **map, interpretability & explainability**

One more D3 lab and one R lab

Your final projects

Final project milestone: **April 30** (or **May 7** if the class prefers).

(Note: if you haven't attended lectures, at least review lecture slides)

User Interface

CMSC471: Introduction to Information Visualization

Fumeng Yang, Assistant Professor

April 15, 2025

University of Maryland, College Park

<https://fumeng-yang.github.io/CMSC471>

Your assignment submissions remind me of...

**The worst volume control
UI in the world**

Advanced Volume Control



Volume Control

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 ☐ 11

☐ 12 ☐ 13 ☐ 14 ☐ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐ 20 ☐ 21

☐ 22 ☐ 23 ☐ 24 ☒ 25 ☐ 26 ☐ 27 ☐ 28 ☐ 29 ☐ 30 ☐ 31

☐ 32 ☐ 33 ☐ 34 ☐ 35 ☐ 36 ☐ 37 ☐ 38 ☐ 39 ☐ 40 ☐ 41

☐ 42 ☐ 43 ☐ 44 ☐ 45 ☐ 46 ☐ 47 ☐ 48 ☐ 49 ☐ 50 ☐ 51

☐ 52 ☐ 53 ☐ 54 ☐ 55 ☐ 56 ☐ 57 ☐ 58 ☐ 59 ☐ 60 ☐ 61

☐ 62 ☐ 63 ☐ 64 ☐ 65 ☐ 66 ☐ 67 ☐ 68 ☐ 69 ☐ 70 ☐ 71

☐ 72 ☐ 73 ☐ 74 ☐ 75 ☐ 76 ☐ 77 ☐ 78 ☐ 79 ☐ 80 ☐ 81

☐ 82 ☐ 83 ☐ 84 ☐ 85 ☐ 86 ☐ 87 ☐ 88 ☐ 89 ☐ 90 ☐ 91

☐ 92 ☐ 93 ☐ 94 ☐ 95 ☐ 96 ☐ 97 ☐ 98 ☐ 99 ☐ 100

☐ Mute

UI design is (probably) **not** the
most important thing for a web
but **quite** important

UI Design

The **art** and **science** of designing user interfaces for digital products

Enhance usability, accessibility, and user satisfaction

Direct influence on user experience and business outcomes

Core UI Design Principles

Usability: Make interfaces easy to use

Consistency: Maintain uniform elements across the interface

Feedback: Provide clear responses to user actions

Hierarchy: Guide the user's attention to what matters most

Clarity: Ensure information is presented clearly

Accessibility: Design for users of all abilities

Bad alignment

Never, ever, ever Center paragraph text

Pellentesque sit amet egestas erat,
vitae posuere magna. Quisque
bibendum quam sed consequat
bibendum. Vestibulum dignissim
quam id ipsum molestie, et
pellentesque quam ornare. Morbi
quis commodo nisi, volutpat gravida
lacus. Integer mollis orci nec nunc
laoreet, sit amet dictum nunc
interdum. Quisque sed laoreet ex.



Pellentesque sit amet egestas erat,
vitae posuere magna. Quisque
bibendum quam sed consequat
bibendum. Vestibulum dignissim
quam id ipsum molestie, et
pellentesque quam ornare. Morbi
quis commodo nisi, volutpat gravida
lacus. Integer mollis orci nec nunc
laoreet, sit amet dictum nunc
interdum. Quisque sed laoreet ex.

Lack of visual hierarchy



Jim Raptis
MagicPattern.design Founder

Followers: 5k **Products:** 3 **Posts:** 353

Follow



Jim Raptis
MagicPattern.design Founder

 **5k** followers  **3** products  **353** posts

Follow

Scale x1.25 the title size to highlight it

Reduce opacity to 50%

Use icons to assist user visually identify context and reduce their opacity to 30% to drive attention away of them

Change layout orientation to reduce useless white-space and leverage F-Pattern (people read from left-to-right shaping an F shape).

Long dropdown

Searchable menu

Group them

Country

Country

|

Afghanistan

Albania

Algeria

Country

So|

Solomon Islands

Somalia

South Africa

Wines

- Select -

Wines

- Select -

Red wines

Cabernet Sauvignon

Merlot

Malbec

White wines

Long dropdown

Searchable menu

Group them

Country

Country

|

Afghanistan

Albania

Algeria

Country

So|

Solomon Islands

Somalia

South Africa

Select catagories

- Select -

Select catagories

Art, Biology, Chemistry

☐ Science

☒ Biology

☐ Physics

☒ Chemistry

☒ Art

Resources and Readings

<https://uxdesign.cc/ui-cheat-sheet-dropdown-field-a30025c0f432>

<https://uxdesign.cc/login-form-ui-design-guide-c5c9c93ae482>

<https://uxdesign.cc/whitespace-in-ui-design-44e332c8e4a>

<https://uxdesign.cc/10-principles-for-typography-usage-in-ui-design-a8f038f43ffd>

<https://uxdesign.cc/ui-design-basics-screens-734bfbeffca9>

All from <https://uxdesign.cc/search?q=ui+design>

Today

Discuss your final project
with your peers

TAs and I will also
circulate in the room and
join the conversations

Remember to fill in the
peer back spreadsheet

first row

Front 

Group 1
Urban/Infra

Group 2
Climate

Group 3
Health

Group 4
Justice

Group 5
Tech/Econ

Group 6
Media

Group 7
Algo/ML

Back 